

KSHITIJ WANKAR

Python & AI Backend Engineer — Retrieval Systems, LLM Agents & MCP Tooling

kishor.kshitij@nmiet.edu.in | github.com/Kshitij1ghk | B.E. Information Technology, 2026 — PCET-NMIT, Pune

SUMMARY

Final-year Information Technology engineering student with a strong foundation in Python, C#/.NET, OOP, backend development, databases, and data structures & algorithms. Hands-on experience building applied GenAI projects involving retrieval-augmented generation, vector search, LLM tool-calling, and Model Context Protocol (MCP). Comfortable building and shipping end-to-end software using REST APIs, FastAPI, and modern AI development workflows.

TECHNICAL SKILLS

Languages: Python (primary — OOP, asyncio, type hints), C#, JavaScript (ES6+), SQL, C/C++ basics

AI / GenAI: Retrieval-augmented generation (RAG), embeddings & vector search, LLM APIs, prompt engineering, AI agents, tool-calling, Model Context Protocol (MCP)

Backend & Data: FastAPI, Flask, ASP.NET Core MVC, Entity Framework Core, RESTful APIs, async programming, Pydantic, MySQL, relational schema design, Qdrant (vector DB, HNSW)

Developer Tools: Git/GitHub, GitHub REST & GraphQL APIs, pytest, Visual Studio, JupyterLab, Vercel

Frontend: HTML5, CSS3, JavaScript, Bootstrap, Tailwind CSS

PROJECTS

Business RAG Assistant — Python, FastAPI, Qdrant, FastEmbed, Gemini

github.com/Kshitij1ghk/business-rag-assistant

- Built an end-to-end retrieval-augmented QA system over internal policy documents: chunking → 384-dim BAAI/bge-small embeddings via local ONNX inference (no external embedding API, so no per-query cost) → Qdrant storage with HNSW and cosine similarity → grounded answer synthesis with source citations, similarity scores and confidence bars in the UI.
- Hardened against the failure modes common to RAG systems: a 0.40 similarity floor returns a deterministic refusal *without* calling the LLM when nothing relevant is retrieved, and retrieved chunks are passed as untrusted data rather than instructions to resist prompt injection. 16 passing pytest tests plus a diagnostic retrieval script for tuning the threshold.

MCP-Powered Developer Assistant — Python, MCP, GitHub API, AST, pytest

github.com/Kshitij1ghk/MCP-Powered-Developer-Assistant

- Designed an MCP client/orchestrator that connects separate code, GitHub and test MCP servers into multi-step agent workflows — locate a bug, inspect symbols, triage the matching issue, generate a patch, and validate it — with interactive [Y/N] approval gates before any change is written.
- Implemented codebase access over Python AST parsing (classes, functions, arguments, docstrings, line ranges) confined to an ALLOWED_DIR jail with path-traversal prevention, plus diff-based patch generation and sandboxed pytest execution with timeouts; 16 unit and integration tests cover the code, patching, GitHub and agent-flow layers.

Asterion Astrologer — Python, Flask, Groq LLM API, Whisper, Vercel

github.com/Kshitij1ghk/Asterion-Astrologer

- Shipped a full-stack AI astrology web app — Flask backend, responsive frontend — generating Vedic and KP birth charts with planetary positions computed to four decimal places via custom Swiss Ephemeris algorithms, plus Vimshottari Dasha periods, Star Lords / Sub Lords analysis and daily horoscopes.
- Added a conversational assistant using Groq-hosted LLMs (Llama 3.3 / GPT-OSS) with conversation memory and quick-action prompts, plus Whisper speech-to-text / text-to-speech for Hinglish and English voice input; birthplace search via OpenStreetMap, deployed on Vercel (asterion-gamma.vercel.app).

EDUCATION

Bachelor of Engineering, Information Technology

Aug 2023 – 2026

PCET-Nutan Maharashtra Institute of Engineering and Technology, Pune | CGPA 8.34/10

Higher Secondary (Class 12), Science

May 2021 – Jul 2022

Shree Mahrishi Vidya Mandir, Chandrapur | 75.2%

Secondary (Class 10)

May 2019 – Jul 2020

St. Michael's English School, Chandrapur, Maharashtra | 93.2%

ADDITIONAL INFORMATION

- Leadership:** Technical Coordinator, Student Association of Information Technology — runs technical events and code sessions for the cohort.
- Certifications:** HTML, CSS and JavaScript for Web Developers — Johns Hopkins University (Coursera); Python Data Structures — University of Michigan (Coursera).
- Interests:** Reading fictional and philosophical genre.